

Aim: Implementing Linear Data structures using Arrays and performing insertion, deletion, copying an array to other, searching and sorting

Diagram for **Insertion** of an element in an array:

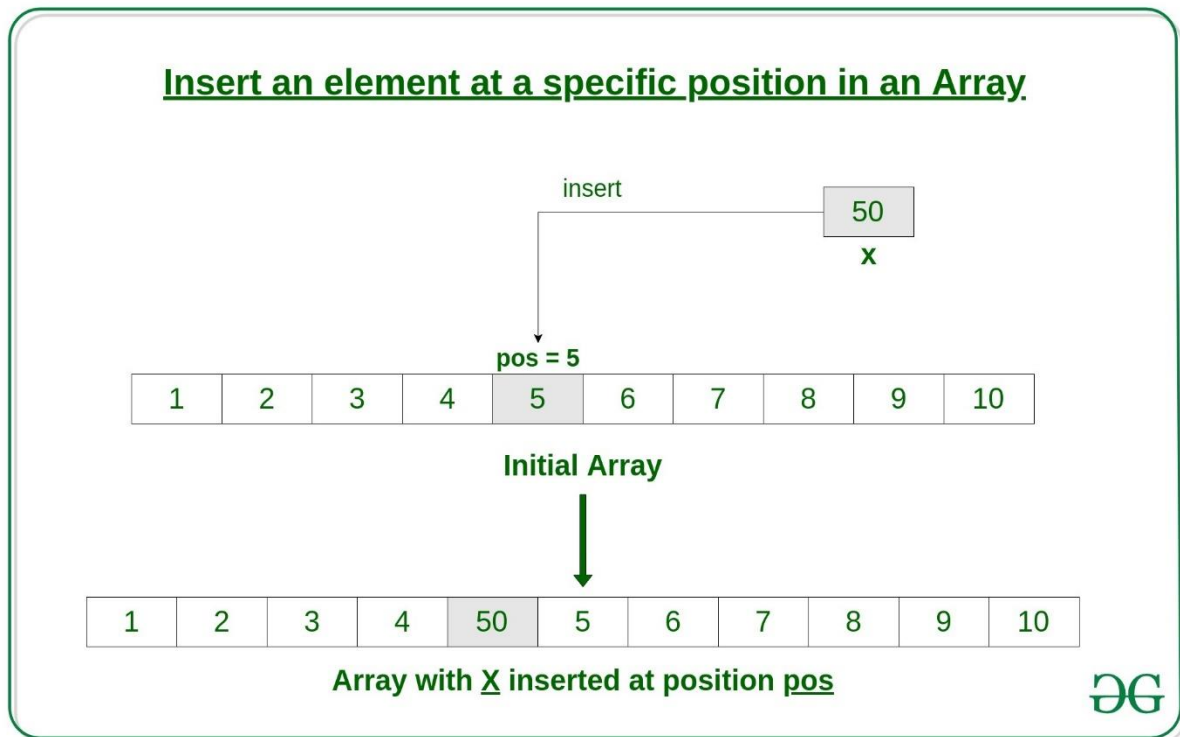


Diagram for **Deletion** of an element from an array:

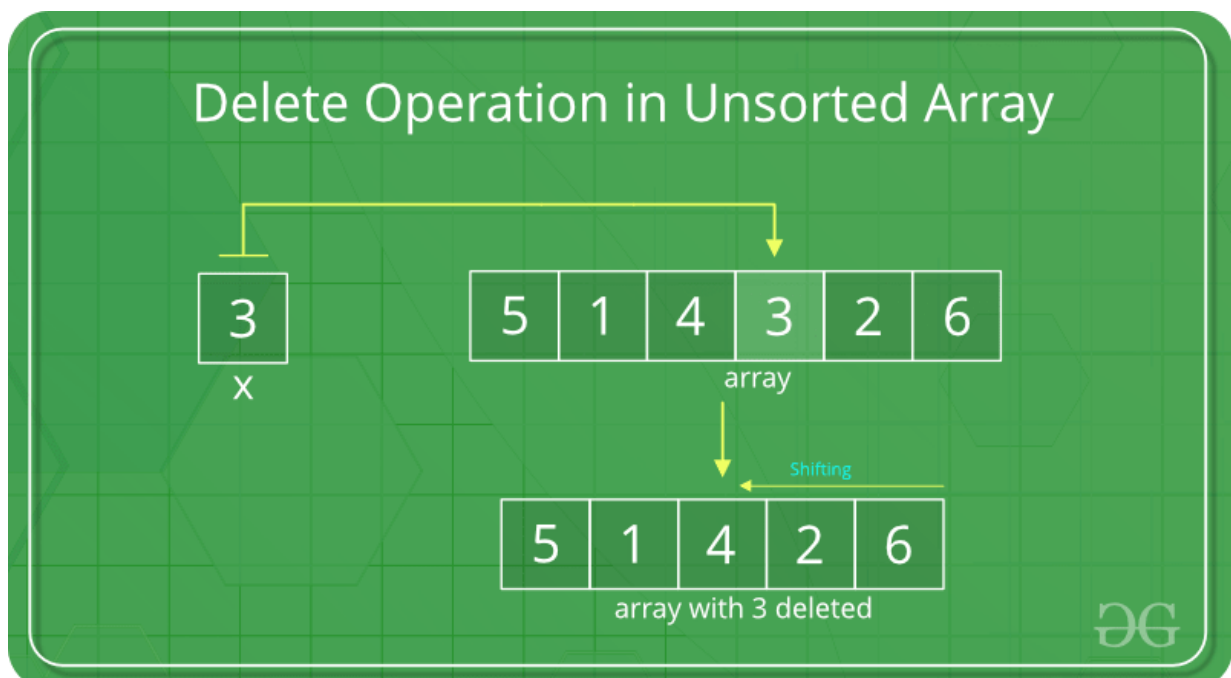


Diagram for implementing **Bubble Sort Algorithm**:

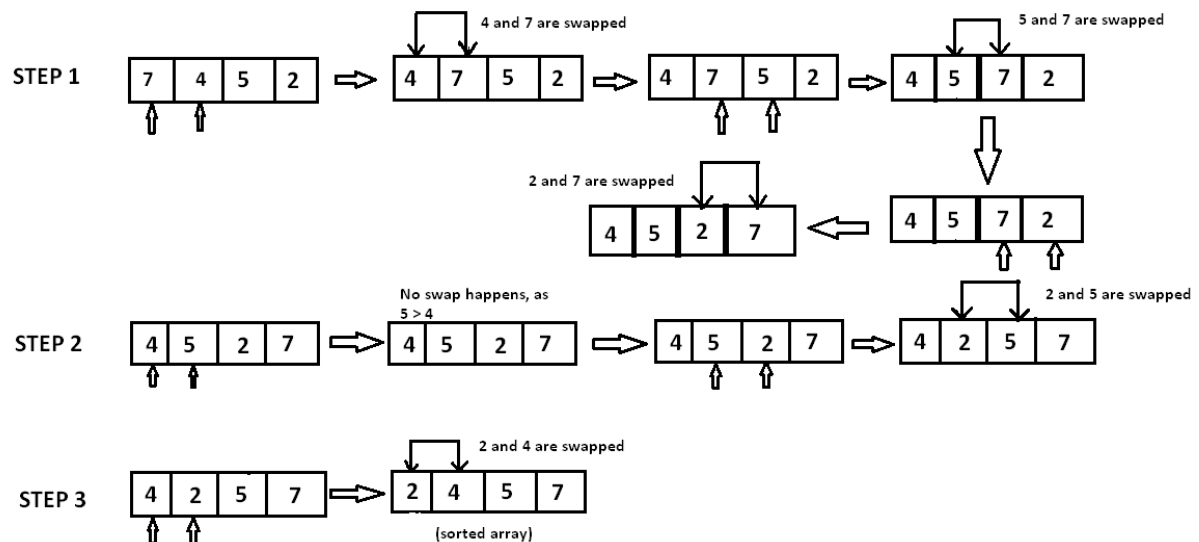
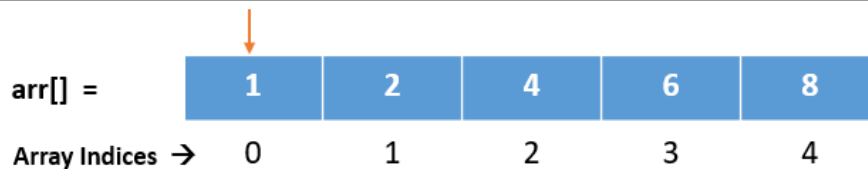


Diagram for implementing **Linear Search Algorithm**:

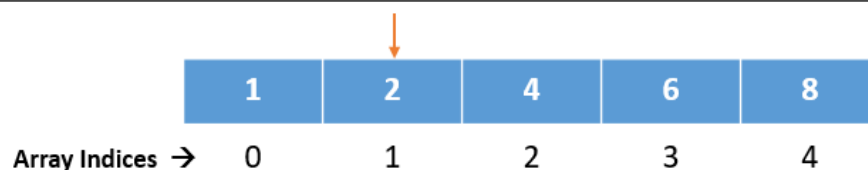
Linear Search Algorithm

Element to search for: 4

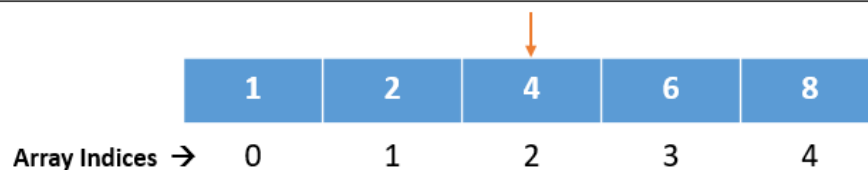
Iteration 1: Begin search from index 0 and check $\text{arr}[0] == \text{element}$



Iteration 2: Move to next element and check $\text{arr}[1] == \text{element}$



Iteration 3: Move to next element and check $\text{arr}[2] == \text{element}$



Element Found!

Program:

```
#include<stdio.h>
void main()
{
    int
    n,m,p,i,j,a[10],b[10],c[10],loc,z,temp,x,found=0;
    //Creating an array
    printf(" Enter the number of elements: ");
    scanf("%d",&n);
    m=n;
    p=n;
    for(i=0;i<n;i++)
    {
        printf("\n Enter number: ");
        scanf("%d",&a[i]);
    }
    //Copying an array
    for(i=0;i<n;i++)
    {
        b[i]=a[i];
        c[i]=a[i];
    }
    //Displaying the original array
    printf("\n The original array:");
    for(i=0;i<n;i++)
    {
        printf("\t %d",a[i]);
    }
    //Displaying the copied array
    printf("\n The copied array:");
    for(i=0;i<n;i++)
    {
        printf("\t %d",b[i]);
    }
    //Deletion of an element
    printf("\n Enter the location of the element to
be deleted: ");
    scanf("%d",&loc);
    while(loc<n)
    {
        a[loc-1]=a[loc];
        loc++;
    }
    n--;
    for(i=0;i<n;i++)
    {
        printf("\t %d",a[i]);
    }
    //Insertion of an element
```

Output:

```
Enter the number of elements: 4

Enter number: 1

Enter number: 3

Enter number: 2

Enter number: 4

The original array:  1   3   2   4
The copied array:   1   3   2   4
Enter the location of the element to be
deleted: 2
          1   2   4
Enter the element to be inserted: 5

Enter the location of the element to be
inserted: 3
          1   3   5   2   4
Sorted Array:
          1   2   3   4
Enter element to be searched: 2
Element is present
```

```

    printf("\n Enter the element to be inserted:
");
    scanf("%d",&z);
    printf("\n Enter the location of the element
to be inserted: ");
    scanf("%d",&loc);
    for(i=m-1;i>=loc-1;i--)
    {
        b[i+1]=b[i];
    }
    b[loc-1]=z;
    for(i=0;i<m+1;i++)
    {
        printf("\t %d",b[i]);
    }
//Bubble Sort
    for(i=0;i<p;i++)
    {
        for(j=0;j<p-i;j++)
        {
            if(c[j]>c[j+1])
            {
                temp=c[j];
                c[j]=c[j+1];
                c[j+1]=temp;
            }
        }
    }
    printf("\n Sorted Array: \n");
    for(i=0;i<p;i++)
    {
        printf("\t %d",c[i]);
    }
//Linear Search
    printf("\n Enter element to be searched: ");
    scanf("%d",&x);
    for(i=0;i<p;i++)
    {
        if(x==c[i])
        {
            printf("\n Element is present");
            found=1;
        }
    }
    if(!found)
    {
        printf("\n Element is not present");
    }
}

```